

Predicting UK Political Party Affiliation from Conference Speeches

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Abstract

This paper checks whether we can predict which political member belongs to which political party based on the speech they have given in a parliamentary conference. By using the UK Political Speeches dataset, we selected two parties, Labour and Conservative, and built a corpus of 240 speeches that cover several decades. We run these speeches through a text classification pipeline that includes domain-specific pre-processing, TF-IDF feature extraction with bi-grams, and three classifiers, which are Logistic Regression, Naive Bayes, and Random Forest. During this process, we also found something interesting: Labour speakers tend to mention Conservative party names when they are criticizing them, which was creating false signals in the model. After we removed this noise, our best model, such as Logistic Regression, reaches 89.6% accuracy and F1 score of 0.895 on the test set. We also looked for the speeches that the model got wrong, and some of them are genuinely interesting, like Theresa May and Margaret Thatcher, whose language actually sounds more like the other party. Overall, data sets make it clear that most political parties have quite the same consistent and strong vocabulary.

1 Introduction

Political language is a very big domain for computational linguistics. Many politicians have developed their own way of speaking, which is shaped by ideology, policy priorities, and rhetorical tradition. This raises the natural question: Is party affiliation encoded strongly enough in language that a machine learning model can detect it automatically?

This paper looks at that question in the context of UK politics, focusing on annual party conference speeches, a genre where politicians speak directly to their own base and ideological language is very prominent. Specifically, we ask: *Can we predict*

whether a conference speech was delivered by a Labour or Conservative MP, by using only the text of the speech?

We approach this as a binary text classification problem. Our pipeline includes domain-specific pre-processing, TF-IDF vectorization with uni-grams and bi-grams, and three supervised classifiers evaluated via stratified cross-validation. One key thing we found during this process is cross-party reference bias, which is basically when Labour speakers frequently name Conservative opponents, and this was causing party-name terms to become false Labour signals, so we had to remove them.

Our best model makes up to 89.6% accuracy, which reflects that party language and tone are both strong and also learnable. Beyond numbers, we also look for the speeches that our model gets wrong. That reveal some historically interesting cases where some political unintentionally or by mistake, start using the language of the opposite party.

2 Related Work

The delectation of political speeches gets a lot of attention lately in computational linguistics and NLP (Natural Language Processing). One of the early work by [Thomas et al. \(2006\)](#) reflect that the voting behavior in the US Congress can be predicted by the floor debate transcripts, which confirms one thing: political affiliation is detectable in legislative language.

In the UK context, [Duthie et al. \(2016\)](#) looked at argumentation mining in parliamentary debate and showed that rhetorical structure is different across parties in a systematic way. More broadly, authorship attribution, which is the task of identifying who wrote a text, shares a lot of overlap with party prediction, because both of them rely on stylistic and lexical consistency ([Stamatatos, 2009](#)).

TF-IDF combined with linear classifiers like Logistic Regression has proven to work well for political text classification. [Yu et al. \(2008\)](#) showed that sparse bag-of-words representations capture ideological signals well, and they often perform better than more complex models when the corpus is small. This is one of the main reasons we chose TF-IDF over dense neural embeddings, given that our dataset only has 240 speeches.

Previous work on UK politician speeches is quite limited, which makes our project one of the first proper classification analyses in this domain. Our work also adds to the existing research by introducing iterative noise removal as a clear methodological step. Moreover, by doing a quantitative misclassifications analysis that is grounded in a historical political context.

3 Data

3.1 Dataset

We had used the UK political speech dataset ([Sale, 2023](#)), which is publicly available and hosted in Kaggle. The data consist of transcripts of political parties' conference speeches that were delivered over the decade in the UK. We use an `speeches_bps.csv` file, which includes speaker name, date, party affiliation, and full speech text, and this is the only file in the dataset that has a party label, which is required for supervised classification.

3.2 Filtering and Class Balance

The full dataset contains speeches from five parties. We restrict our analysis to Labour and Conservative speeches, and there are two reasons for this. First, only these two parties have enough speeches for reliable supervised learning, with 123 Labour and 117 Conservative speeches, respectively. Second, binary classification makes it easier to interpret features and do more meaningful analysis of a vocabulary that separates two parties. The final corpus has 240 speeches with a class balance of 95.1%, where 100% would be a perfect balance.

3.3 Corpus Characteristics

Speeches vary quite a lot in length, with an average of 2,916 words before pre-processing. The corpus covers several decades of UK political history, which means there is potential temporal variation in language use. This is both a limitation because vocabulary shifts over time may confuse party sig-

nals, and also an opportunity, because historically significant speeches give us interpretable misclassification cases which we had discussed in Section 5.

3.4 Limitations

The dataset has only conference speeches, which makes it stronger as it is fully focused on the type of speeches the party gave. Unlike parliamentary speeches or any media interview. Moreover, the data set doesn't contain any speaker demographic information, which means that we can't look at confounding variables like the speaker's gender or region.

4 Methodology

4.1 Preprocessing

Each speeches was gone threw a 6-step pipeline. First, we lowercase the text and remove all the non - alphabetic characters. Tokenisation was done using NLTK's `word_tokenize`. Stop words were removed using a combined list that includes NLTK's standard English stop words and a custom domain-specific list of 52 political speech terms, which includes transcription artifacts like *applause* and *laughter*, parliamentary courtesy terms like *minister* and *honourable*, and universal conference filler words like *conference* and *delegates*. Tokens were then lemmatised using NLTK's `WordNetLemmatizer`, and any tokens of two characters or fewer were discarded. This pipeline retained an average of 51.3% of words per speech, which is within our target range of 40–60%.

4.2 Iterative Noise Removal

After running the model for the first time, we looked at the top discriminating features that Logistic Regression identified. This is where we found cross-party reference bias, which is basically that Labour speakers frequently name Conservative opponents like *tories* and *Tory* when criticising them, and this was causing these terms to appear as false Labour signals. Party name terms like *conservative*, *labour*, *tory*, and *tories* were added to the stopword list, along with contraction residuals like *weve* and *theyre*, transcription artifacts like *cheer* and *cheers*, and formal address terms like *gentlemen* and *madam*. The final stop word list had 281 terms in total. After rerunning the pipeline with this updated list, Logistic Regression accuracy improved from 87.5% to 89.6%, which confirmed that the noise removal was actually helping.

4.3 Feature Extraction

We applied TF-IDF vectorization with the vocabulary capped at 5,000 features, using uni-grams and bi-grams (`ngram_range=(1,2)`). Bi-grams are useful because they capture politically meaningful phrases like *minimum wage* and *free enterprise* that don't really make sense when you split them into individual tokens. A minimum document frequency of 2 (`min_df=2`) was used to exclude very rare words, and sublinear TF scaling (`sublinear_tf=True`) was applied so that longer speeches do not just dominate because they have more words. The vectoriser was fitted only on the training data to prevent data leakage.

4.4 Classifiers

We trained and compared three classifiers. **Logistic Regression** (LR) is a linear classifier that works well with sparse, high-dimensional text data, and its coefficients are directly interpretable, which means we can see which terms most strongly predict each party (Yu et al., 2008). **Naive Bayes** (NB) works as the probabilistic baseline, and even though it assumes features are independent, it still performs well on small text corpora (Stamatatos, 2009). **Random Forest** (RF) is an ensemble method that aggregates 200 decision trees and can capture non-linear feature interactions that linear models cannot.

4.5 Evaluation

With the help of stratified 5-fold cross-validation, we evaluated all the models on the training set, which consists of 192 speeches. Moreover, it also held out a test set of 48 speeches, which was produced with the distribution of 80/20 stratified split with `random_state=42` for reproducibility. We also report the macro-averaged F1 score as a main metric with accuracy. Cross-validation gives a more reliable performance estimate than a single split, especially given the modest corpus size (Thomas et al., 2006).

5 Results and Findings

5.1 Model Performance

Table 1 shows the final results for all three classifiers after iterative noise removal. Logistic Regression gives the best test performance with 89.6% accuracy and a macro F1 score of 0.895, which shows us that party-distinctive language is strongly learnable from conference speech text. Naive Bayes also performs well at 87.5% accuracy. Random Forest,

on the other hand, even though it had the highest cross-validation F1 of 0.880, gives the weakest test performance at 83.3%, which suggests it is overfitting a little bit to the training data.

Model	CV F1	Test Acc	Test F1
Logistic Regression	0.863	89.6%	0.895
Naive Bayes	0.847	87.5%	0.875
Random Forest	0.880	83.3%	0.833

Table 1: Final model comparison after iterative noise removal. CV F1 is macro-averaged over 5 folds.

Figure 1 shows normalized confusion matrices for all three models. Logistic Regression correctly classifies 20 of 23 Conservative speeches and 23 of 25 Labour speeches on the test set.

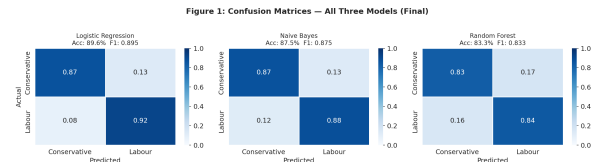


Figure 1: Normalized confusion matrices for all three classifiers on the held-out test set (48 speeches).

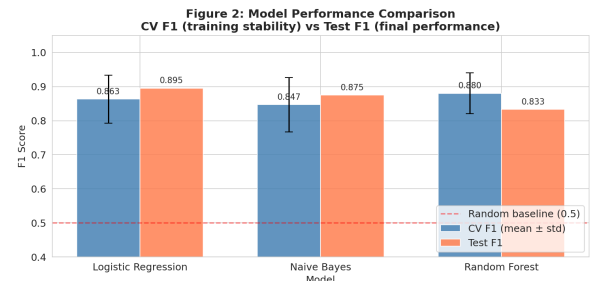


Figure 2: Model comparison: cross-validation F1 versus test F1. The dashed line indicates the random baseline (0.5).

5.2 Discriminating Vocabulary

Figure 3 shows the top 20 most discriminating terms per party that were identified by Logistic Regression coefficients after noise removal. Labour-predictive terms are mostly around social justice themes like *poverty*, *investment*, *minimum wage*, *fairness*, *full employment*, and *antisocial behavior*. Conservative-predictive terms are more around economic and national themes like *empire*, *free enterprise*, *tariff*, *savings*, *nation*, and *prosperity*. This matches quite well with what we know about Labour and Conservative ideological traditions.

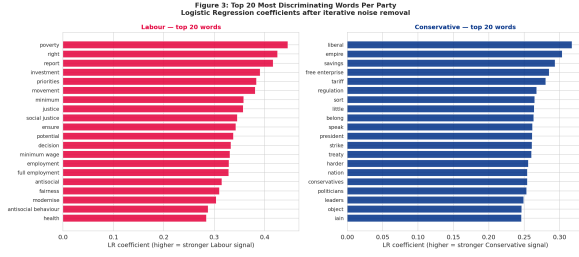


Figure 3: Top 20 discriminating terms per party by Logistic Regression coefficient magnitude after iterative noise removal.

5.3 Misclassification Analysis

Out of 48 test speeches, 5 were misclassified, which is a 10.4% error rate. One important thing to note is that all five errors happened at low model confidence between 0.53 and 0.63, compared to the mean confidence of 0.616 for correct predictions. Figure 4 shows this distribution. This pattern suggests that the errors are coming from genuine linguistic ambiguity rather than the model failing systematically.

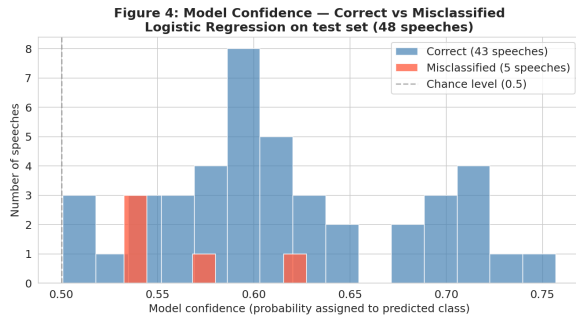


Figure 4: Model confidence distribution for correct versus misclassified speeches. Misclassified speeches cluster near the chance level (0.5).

The misclassifications actually reveal some historically interesting cases. Theresa May’s 2016 and 2017 conference speeches were both predicted as Labour, because she was using very social-justice type framing, talking about burning injustices” and policies for people who are just about managing,” which is language that sounds more like Labour. Margaret Thatcher’s January 1990 speech, which was delivered to the children’s welfare charity, was also misclassified, and this makes sense because the subject matter was much softer than what you would normally expect from a Conservative leader. Ramsay MacDonald’s 1924 speech was predicted as Conservative, and this is interesting because, as the first ever Labour Prime Minister, MacDonald was deliberately using moderate and establishment-

friendly language to show that he could govern credibly. Michael Foot’s 1981 speech was also misclassified because his formal classical oratorical style was quite different from the typical Labour vocabulary of that time period. These cases show that the model has genuinely learned party-linguistic norms, and when politicians deviate from those norms, it corresponds to things that actually happened historically.

6 Conclusion and Future Work

This study shows that UK political party affiliation can be predicted from conference speech text with pretty high accuracy. Our best model, which is Logistic Regression with TF-IDF bigram features, reaches 89.6% accuracy and an F1 score of 0.895 on a held-out test set of 48 speeches, which is well above a random baseline of 50%. These results confirm that party-distinctive vocabulary is strong, consistent, and learnable across several decades of UK political history.

Beyond just the classification performance, this study makes two analytical contributions. First, we identify and document cross-party reference bias as a methodological concern in political text classification, because opposition party names appear disproportionately in attacking party speeches, and this has created false lexical signals that need to be explicitly removed. Second, the qualitative misclassifications analysis reveals historically grounded cases of genuine linguistic ambiguity, which are politicians who deliberately or accidentally adopted the rhetorical style of the opposing party, including Theresa May, Margaret Thatcher, and Ramsay MacDonald.

There are a few limitations that should be noted. The corpus is restricted to conference speeches, a genre where ideological language is particularly prominent, so results may not generalize to parliamentary debate or media interviews. The test set of 48 speeches is quite modest, and performance estimates carry some uncertainty because of that. Lemmatization was applied with a uniform verb POS tag, which may reduce accuracy for nominal forms. Also, temporal variation in political language across decades is not explicitly modeled.

Future work could address these limitations in several ways. Expanding the corpus to include parliamentary Hansard transcripts would help test generalization across speech genres. Adding temporal modeling, for example, decade-stratified analysis,

could reveal how party language has shifted over time. Transformer-based models like BERT (Devlin et al., 2019) fine-tuned on political text may capture semantic nuance that is beyond what bag-of-words representations can do. Finally, extending the classification to include Liberal Democrat speeches, or framing the task as outlier detection to identify potential political defectors, would be a natural extension of his work.

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